Industrial Safety Processes

Safeguarding

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Safeguarding

- Any means of preventing contact between a person and the moving parts of machinery or with equipment that could potentially cause physical harm
- May include:
  - Substitution
  - Administrative controls (i.e., yellow lines on floors)
  - Warning bells and lights
- NOTE: ***machine guarding*** is often a term used only for the application of physical, solid guards on equipment, not administrative systems

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Device (in Safeguarding)

- A mechanism or control designed for safeguarding at the point of operation
- May include:
  - Presence sensing devices
  - Movable barriers
  - Holdout or restraint systems
  - Dual action activation

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- Insert Videos:

- https://youtu.be/IbpPvCLAy8Y
- https://youtu.be/sXMlD8msXgE
- https://youtu.be/5oaXH_1PbRM

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Enclosure

- Safeguarding by fixed physical barriers mounted on or around a machine to prevent access to the moving parts.

- Enclosures are most effective when designed as part of the machine or to be bolted or welded to the frame or the floor.

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Location

- Safeguarding by location results when a hazard is physically inaccessible under normal operating conditions or use.
- Very limited safeguarding technique
- Should only be used if other systems not possible

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Pinch Point

- Any place where a body part can be caught between two or more moving mechanical parts, or one fixed and one moving part in the same plane
- Belt and pulley systems
- Spring loaded systems
- May also referred to as “nip points”

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Nip Points or Bites

- A hazardous area created by two or more mechanical parts rotating in opposite directions within the same plane and in close interaction
- Rollers
- Belt and pulley systems
- May also be referred to as “pinch points”

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Point of Operation

- The area on a machine where material is positioned for processing or where work is actually being performed on the material

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Fencing

- Safeguarding by means of a locked fence or rail enclosure that restricts machine access to authorized personnel
- Common for automated machinery, such as elevator drives or robots
- Not ideal safeguarding technique – should only be used when other options not practical

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Point of Operation

- Main goal is to keep worker away from moving parts of machinery during operation
- Need to find balance between permitting access for materials while keeping worker body parts from being machined
- Can be direct guards, sensing devices, pull-backs, two hand controls

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Shear Point

- A hazardous area created by the cutting movement of a mechanical part past a stationary point on a machine
- Paper cutters
- Metal shears

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Power Transmission

- Includes all mechanical parts – such as gears, cams, shafts, pulleys, belts, clutches, brakes, and rods – that transmit energy and motion from source of power to equipment or a machine

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Openings for Safeguarding

- 3/8th inch or 9 mm is considered to be “relatively safe”, **BUT**

- **¼ inch or 6 mm** to prevent finger access

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Main Power Transmission Areas

- Rotating, reciprocating, and transverse motions
- In-running nip points
- Cutting actions
- Punching, shearing, and bending actions

- Could be confused with point of operation

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Hazardous Energy

- Could be any form of stored energy, including:
  - Electrical
  - Chemical
  - Gravity
  - Steam
  - Hydraulic
  - Pneumatic
  - Thermal

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Lockout/Tagout

- Isolating energy sources is a major issue when engaged in maintenance on systems
- Commonly associated with electrical work, lockout/tagout applies to the isolation of any energy source that may harm workers during maintenance activities
- Should be managed by a well-defined program and procedures

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Interlocking Devices

- Device that must be in either a closed or open position before a machine will start
- Can be mechanical or electrical trigger or sensor
- Could cut power to the system or simply provide a warning (seat belt buzzer)

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Some Problems

- Easy removal of guards or maintenance
  - Some jurisdictions do not permit “quick release” mechanisms or guards, require “bolted in place”
- Hard to removal/reinstall may mean that guards are not replaced after maintenance
- Some maintenance may not get done with guards in place
- Guarding is not just for point of operation

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The Creative Worker

- If safeguards are perceived as slowing a worker down in his/her task, the worker is likely to attempt to defeat the safeguard
- Some people will expend more time and energy looking for ways around a guard, or defeating it completely, than any time or energy savings realized

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